Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

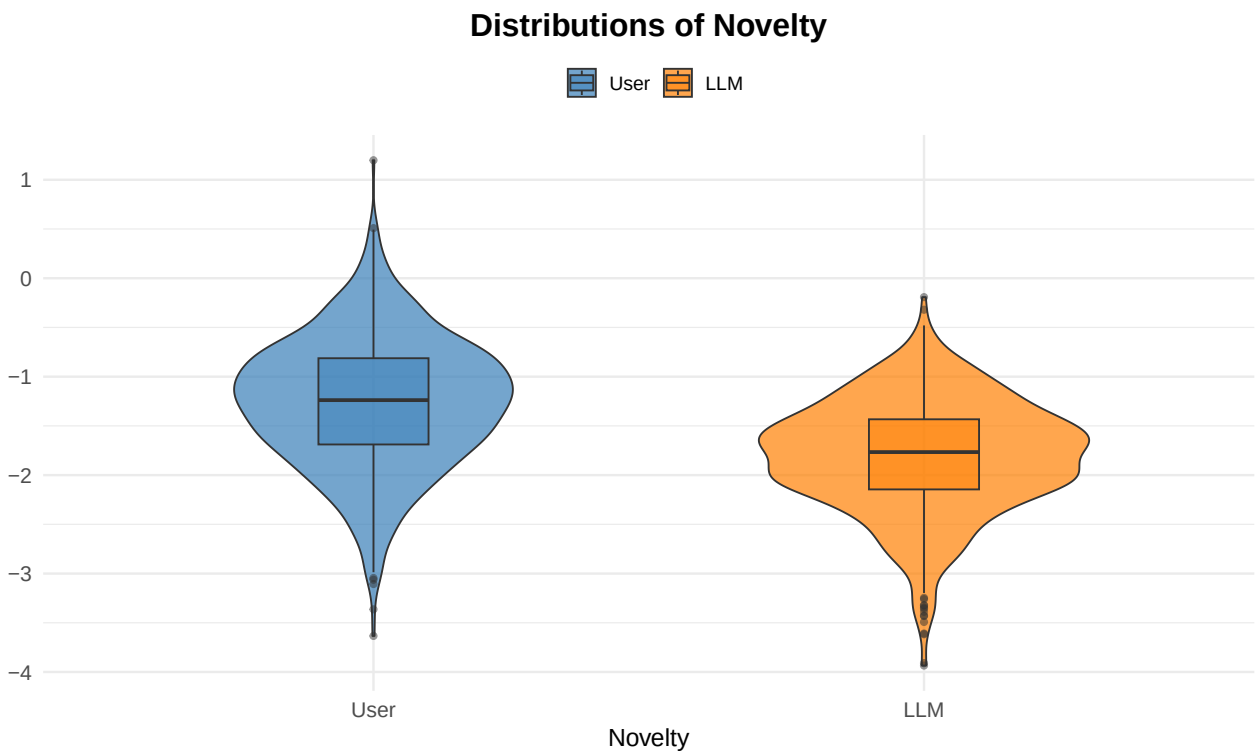
1.1 1.1 Load Data

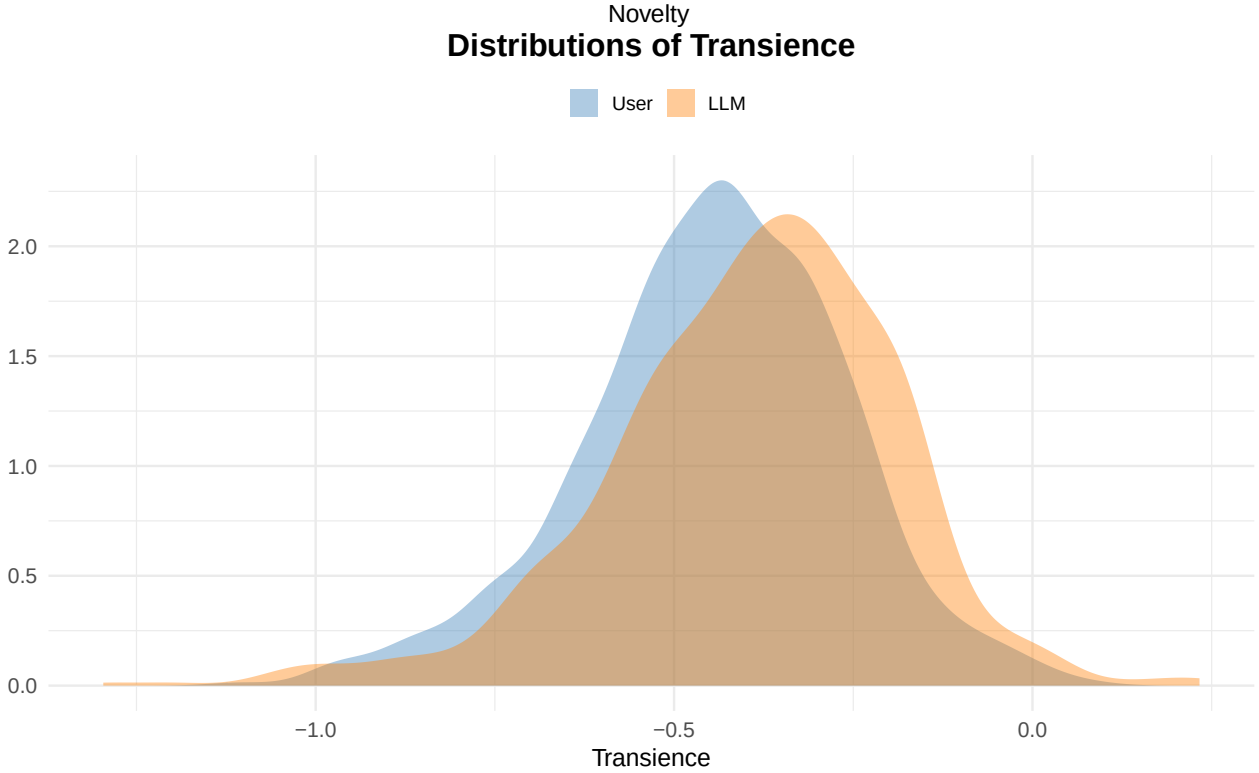
1.2 1.2 Reshape Data

1.3 1.3 Descriptive Statistics

Table 1: Descriptive Statistics for Novelty Metrics by Agent

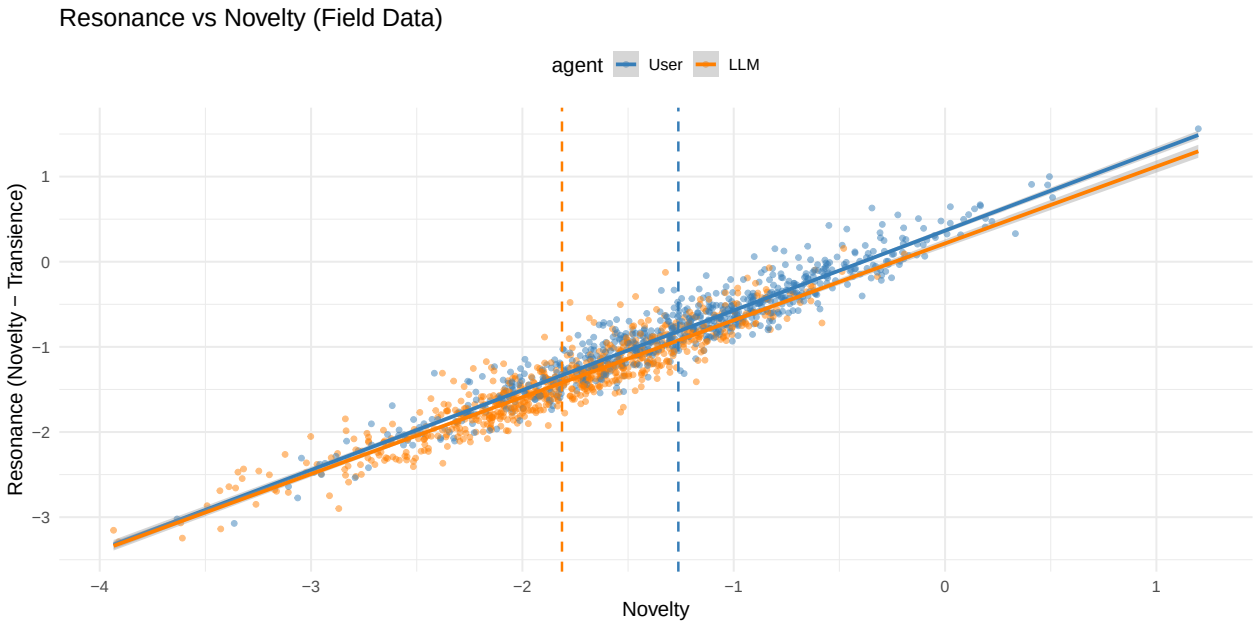
agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	700	-1.262	0.668	-0.445	0.183	-0.817	0.651
LLM	700	-1.813	0.578	-0.391	0.201	-1.422	0.556



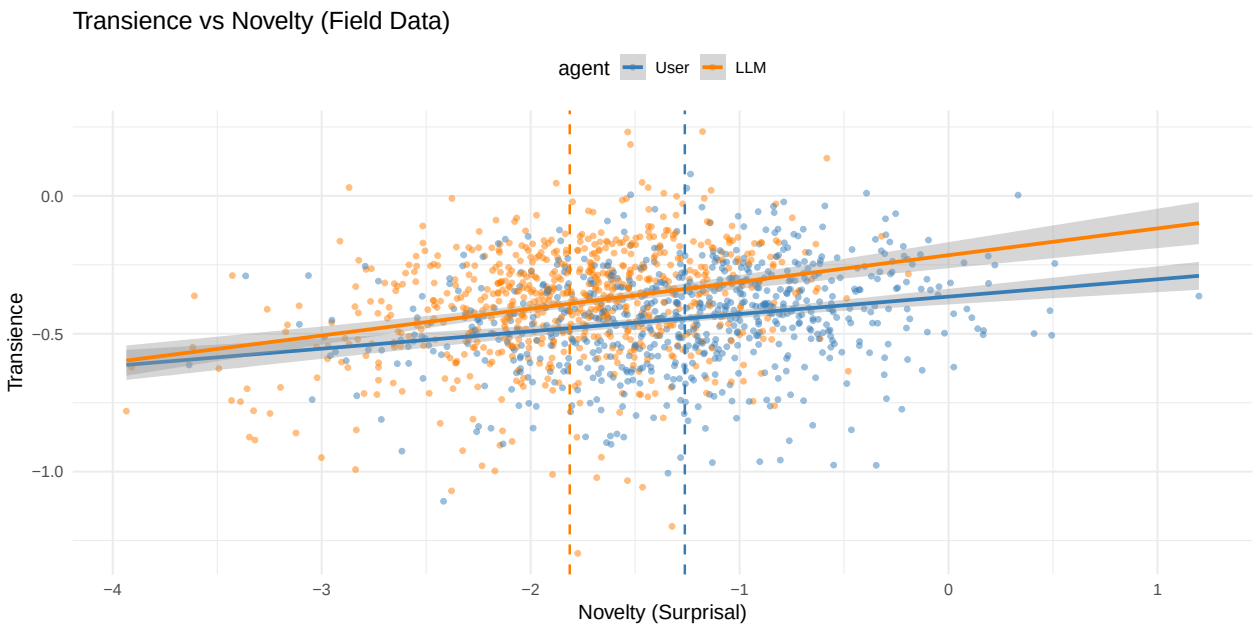


2 2. Novelty and Resonance Analysis

2.1 2.1 Resonance vs Novelty Plot



2.2 2.2 Transience vs Novelty Plot



3 3. Statistical Models

3.1 3.1 Resonance Model

```
## === Resonance Model ===

##                     Estimate Std. Error        df   t value     Pr(>|t|)
## (Intercept)        0.38083765 0.01578280  881.3152 24.129921 3.629465e-99
```

```
## surprise           0.94923119 0.01054567 1395.9622 90.011432 0.000000e+00
## agentLLM          -0.13969671 0.02744131 1391.3007 -5.090745 4.054814e-07
## surprise:agentLLM -0.03206004 0.01601400 1392.4157 -2.002002 4.547824e-02

##
## === Slope Comparison (Resonance) ===

##  agent surprise.trend     SE   df lower.CL upper.CL
##  User           0.949 0.0106 1396    0.929    0.970
##  LLM            0.917 0.0123 1389    0.893    0.941
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3.2 3.2 Transience Model

```
## === Transience Model ===

##                      Estimate Std. Error        df    t value     Pr(>|t|)
## (Intercept)       -0.38083765 0.01578280  881.3151 -24.129921 3.629507e-99
## surprise           0.05076881 0.01054567 1395.9622   4.814184 1.638784e-06
## agentLLM           0.13969671 0.02744131 1391.3007   5.090745 4.054816e-07
## surprise:agentLLM  0.03206004 0.01601400 1392.4157   2.002001 4.547824e-02

##
## === Slope Comparison (Transience) ===

##  agent surprise.trend     SE   df lower.CL upper.CL
##  User          0.0508 0.0106 1396   0.0301   0.0715
##  LLM           0.0828 0.0123 1389   0.0587   0.1070
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4 4. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Pacific/Auckland
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.1   here_1.0.2      afex_1.4-1      cowplot_1.1.3
##  [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
```

```
##  [9] lmerTest_3.1-3  lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
## [17] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] TH.data_1.1-2       pacman_0.5.1        digest_0.6.37
##  [7] timechange_0.3.0    estimability_1.5.1  lifecycle_1.0.4
## [10] survival_3.7-0      magrittr_2.0.3      compiler_4.4.2
## [13] rlang_1.1.4         tools_4.4.2         utf8_1.2.4
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.0.5           plyr_1.8.9          abind_1.4-8
## [22] multcomp_1.4-26     withr_3.0.2         numDeriv_2016.8-1.1
## [25] grid_4.4.2          fansi_1.0.6         xtable_1.8-4
## [28] colorspace_2.1-1    scales_1.3.0        MASS_7.3-61
## [31] tinytex_0.52        cli_3.6.3           mvtnorm_1.3-3
## [34] rmarkdown_2.28      crayon_1.5.3        ragg_1.3.2
## [37] generics_0.1.3      rstudioapi_0.16.0   reshape2_1.4.4
## [40] tzdb_0.4.0          minqa_1.2.8         splines_4.4.2
## [43] assertthat_0.2.1    parallel_4.4.2      vctrs_0.6.5
## [46] boot_1.3-31         sandwich_3.1-1      carData_3.0-5
## [49] car_3.1-3           hms_1.1.3           pbkrtest_0.5.3
## [52] bit64_4.0.5         Formula_1.2-5       systemfonts_1.1.0
## [55] glue_1.7.0          nloptr_2.1.1        codetools_0.2-20
## [58] gtable_0.3.5        munsell_0.5.1       pillar_1.9.0
## [61] htmltools_0.5.8.1   R6_2.5.1            textshaping_0.4.0
## [64] rprojroot_2.1.1     vroom_1.6.5         evaluate_1.0.3
## [67] lattice_0.22-6      backports_1.5.0     broom_1.0.6
## [70] Rcpp_1.0.13         coda_0.19-4.1       nlme_3.1-166
## [73] mgcv_1.9-1          xfun_0.54           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.